Pune Vidyarthi Griha's

College of Engineering Technology & Management, Pune

Department of Artificial Intelligence & Data Science

Instructions for writing BE Project Work - I Report

1. Prepare two copies of your project report in latex.
2. Limit your project report 40 to 50 **pages (Excluding certificate, Table of Contents, acknowledgement, glossary and references)**
3. The footer "**Department of AI & DS, PVGCOET&M 2026-27**" should be included. (10, Times Roman, Center justified)
4. Main part of project report should be **Times Roman, 12 pts, justified** and **1.5 line spacing**.
5. Use paper size **8.5" x 11"** or **A4** (210 x 197 mm). Follow following margins.

Margin	Paper 8.5" x 11"	Paper A4
Top	1"	23.4 mm
Left	1.5"	37 mm
Bottom	1.25"	32 mm
Right	1 "	23.4 mm

6. There should be **double line space** between paragraphs. Each paragraph will begin with a **five-space indentation.**
7. Chapter **titles** should be **bold** with **14 pts** typed in **CAPITAL** letters and should be aligning at the **center** of the page.
8. **Section heading** should be aligning at the **left** with **12 pts** and **bold** and **capitalized**.
9. Section **subheading** should be aligning at the **left** with **title case.**
10. Leave **two** spaces between section heading and **one** space between two-section subheadings.
11. Illustrations (Charts, drawings, photographs and figures.) must be embedded in text.
    - Illustrations must be sharp, clear, black and white**.**
    - Illustrations should not be more than 2 per page. (One is ideal).
    - **Figure No** and title at bottom with **12** pt.
    - **Legends** below the title in **10** pt.
    - Proper margins from all sides.

- Illustrations **should not** be photocopied.

12. Number the pages **centrally** below the footer.
13. References should be in the order as they **appear in the report.**
14. Symbols and notations if any should be included in nomenclature section only.
15. Order of report: -
    - **Cover page** and **front page** **[Refer P1]**
    - **Certificate** from Institute. **[Refer P2]**
    - **Certificate** from industry, if any.
    - Acknowledgement
    - Abstract
    - Table of Contents **[Refer P3]**
    - Chapters
    - References
16. All chapters, section heading and sub headings should be numbered. For chapter use numbers 1, 2,3 and for sub headings 1.1,1.2, 1.3. And section subheadings 1.1.1, 1.1.2 etc.
17. Use the template provided for writing cover page and certificate with the same font size, type and spacing.
18. Page numbers - Start numbering from introduction with 1, other pages like certificate, acknowledgement, abstract, table of contents etc. with small Roman.
19. Plagiarism check report should be attached.

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A PRELIMENERY REPORT ON (12/bold, upper case)

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YOUR PROJECT TITLE GOES HERE (16/bold, upper case)

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SUBMITTED TO THE SAVITRIBAI PHULE PUNE UNIVERSITY, PUNE
IN THE PARTIAL FULFILLMENT OF THE REQUIREMENTS
FOR THE AWARD OF THE DEGREE (12, upper case)

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OF (12, upper case)

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BACHELOR OF ENGINEERING
(ARTIFICIAL INTELLIGENCE & DATA SCIENCE)
(14/bold, upper case)

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SUBMITTED BY

(12, upper case)

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STUDENT NAMES Exam No :(12, bold/upper case)

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DEPARTMENT OF ARTIFICIAL INTELLIGENCE & DATA SCIENCE
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44, VIDYANAGARI, PARVATI, PUNE 411009(12, bold/upper case)

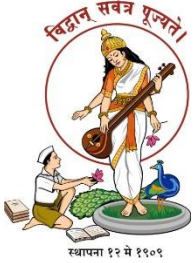
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CERTIFICATE (14, bold/upper case)

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Submitted by
(One blank space)

STUDENT NAMES CRN No : 210808(12, bold/upper case)
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are bonafide students of this institute and work has been carried out by them under the supervision of **Prof. Guide Name** and it is approved for the partial fulfillment of the requirement of Savitribai Phule Pune University, for the award of the degree Bachelor of Engineering(Artificial Intelligence & Data Science).

(12, Sentence case)*(One blank space)*

Prof. Guide Name	**Dr. Mrs. M. P. Atre**
Internal Guide	H.O.D
Dept. of AI&DS.	Dept. of AI&DS.
(12, Sentence case)	

(Dr. S.R.Patil)
I/C Principal,
PVG's College of Engineering Technology & Management, Pune – 09

Place : Pune
Date :

ACKNOWLEDGEMENT

(14, bold/upper case)

It gives us great pleasure in presenting the preliminary project report *on '**BE PROJECT TITLE'**.*

We would like to take this opportunity to thank our internal guide **Prof. Guide Name** for giving us all the help and guidance we needed. We are really grateful for her kind support. Her valuable suggestions were very helpful.

We are also grateful to Dr. Mrs. M. P. Atre, Head-Department of Artificial Intelligence & Data Science , Pune Vidyarthi Griha's College Of Engineering Technology & Management for indispensable support, suggestions. (12 sentence case)

Student Name1
Student Name2
Student Name3
Student Name4
(B.E. AI&DS)

ABSTRACT (14, bold, uppercase)

Statement goes here (12, Sentence case)
Abstract should include 250 to 300 words

Overview of Context of the project, Problem, Solution & Conclusion

.

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LIST OF ABBREVATIONS (FORMAT)

ABBREVIATION	ILLUSTRATION
VPN	Virtual Private Network
IP	Internet Protocol
IDS	Intrusion Detection System
TCP	Transmission Control Protocol

LIST OF FIGURES (FORMAT)

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CHAPTER 1
INTRODUCTION

1.1 PROJECT IDEA

- Idea of project (approx 30 words)

1.2 MOTIVATION OF THE PROJECT

- Motivation of the Project (approx 60 to 70words)

CHAPTER 2
LITERATURE SURVEY

2.1 LITERATURE SURVEY

- Review of the papers, Description, Mathematical Terms ,methodologies used , advantages, limitations(20-25 papers) with respect to project

Paper -1 Title:

Description, Mathematical Terms, methodologies used , advantages, limitations.

Paper 2 Title :

Description, Mathematical Terms, methodologies used , advantages, limitations

…….
……
……
…….

CHAPTER 3

PROBLEM DEFINITION AND SCOPE

3.1 PROBLEM STATEMENT

Description of Problem

3.1.1 Goals and objectives

Goals and Objectives:

- Overall goals and objectives of software, input and output description with necessary syntax, formats described

3.1.2 Statement of scope

- A description of the software with Size of input, bounds on input, input validation, input dependency, Major inputs, and outputs are described without regard to implementation detail.
- The scope identifies what the product is and is not, what it will and won't do, what it will and won't contain.

3.2 SOFTWARE CONTEXT

- The business or product line context or application of the software is to be given

3.3 MAJOR CONSTRAINTS

- Any constraints that will impact the manner in which the software is to be specified, designed, implemented or tested are not adhere.

3.4 METHODOLOGIES OF PROBLEM SOLVING AND EFFICIENCY ISSUES

- The single problem can be solved by different solutions. This considers the performance parameters for each approach. Thus considers the efficiency issues.

3.5 SCENARIO IN WHICH MULTI-CORE, EMBEDDED AND DISTRIBUTED COMPUTING USED

- Explain the scenario in which multi core, embedded and distributed computing methodology can be applied.

3.6 OUTCOME

- Outcome of the project

3.7 APPLICATIONS

- Applications of Project

3.8 HARDWARE RESOURCES REQUIRED

Sr. No.	Parameter	Minimum Requirement	Justification
1	CPU Speed	2GHz	Remark Required
2	RAM	3GB	Remark Required

Table 3.1: Hardware Requirements

3.9 SOFTWARE RESOURCES REQUIRED

Platform:

1. Operating System:
2. IDE:
3. Programming Language:

CHAPTER 4

PROJECT PLAN

4.1 PROJECT ESTIMATES

4.1.1 Reconciled Estimates

4.1.1.1 Cost Estimate

4.1.1.2 Time Estimates

4.1.2 Project Resources

Project resources [People, Hardware, Software, Tools and other resources] based on Memory Sharing, IPC, and Concurrency derived using appendices to be referred.

4.2 RISK MANAGEMENT W.R.T. NP HARD ANALYSIS

This section discusses Project risks and the approach to managing them.

4.2.1 Risk Identification

For risks identification, review of scope document, requirements specifications and schedule is done. Answers to questionnaire revealed some risks. Each risk is categorized as per the categories mentioned in[**?**]. Please refer table 4.1 for all the risks. You can refer following risk identification questionnaire.

1. Have top software and customer managers formally committed to support the project?
2. Are end users enthusiastically committed to the project and the system/product to be built?
3. Are requirements fully understood by the software engineering team and its customers?
4. Have customers been involved fully in the definition of requirements?
5. Do end-users have realistic expectations?

6. Does the software engineering team have the right mix of skills?
7. Are project requirements stable?
8. Is the number of people on the project team adequate to do the job?
9. Do all customer/user constituencies agree on the importance of the project and on the requirements for the system/product to be built?

4.2.2 Risk Analysis

The risks for the Project can be analyzed within the constraints of time and quality

ID	Risk Description	Probability	Impact		
			Schedule	Quality	Overall
1	Description1	Low	Low	High	High
2	Description2	Low	Low	High	High

Table 4.1:Risk Table

Probability	Value	Description
High	Probability of occurrences	>75%
Medium	Probability of occurrences	26−75%
Low	Probability of occurrences	<25%

Table 4.2: Risk Probability definitions [**?**]

Impact	Value	Description
Very high	>10%	Schedule impact or Unacceptable equality
High	5−10%	Schedule impact or Some parts of the project have low quality
Medium	<5%	Schedule impact or Barely noticeable degradation inequality Low Impact on schedule or Quality can be incorporated

Table 4.3: Risk Impact definitions [**?**]

4.2.3 Overview of Risk Mitigation, Monitoring, Management

Following are the details for each risk.

Risk ID	1
Risk Description	Description1
Category	Development Environment.
Source	Software requirement Specification document.
Probability	Low
Impact	High
Response	Mitigate
Strategy	Strategy
Risk Status	Occurred

Risk ID	2
Risk Description	Description2
Category	Requirements
Source	Software Design Specification documentation review.
Probability	Low
Impact	High
Response	Mitigate
Strategy	Better testing will resolve this issue.
Risk Status	Identified

Risk ID	3
Risk Description	Description3
Category	Technology
Source	This was identified during early development and testing.
Probability	Low
Impact	Very High
Response	Accept
Strategy	Example Running Service Registry behind proxy balancer
Risk Status	Identified

4.3 PROJECT SCHEDULE

4.3.1 Project task set

Major Tasks in the Project stages are:

- Task 1:
- Task 2:
- Task 3:
- Task 4:
- Task 5:

4.3.2 Task network

Project tasks and their dependencies are noted in this diagrammatic form.

4.3.3 Timeline Chart

A project timeline chart is presented. This may include a time line for the entire project. Above points should be covered in Project Planner as Annex C and you can mention here Please refer Annex C for the planner

4.4 TEAM ORGANIZATION

The manner in which staff is organized and the mechanisms for reporting are noted.

4.4.1 Team structure

The team structure for the project is identified. Roles are defined.

4.4.2 Management reporting and communication

Mechanisms for progress reporting and inter/intra team communication are identified as per assessment sheet and lab timetable.

CHAPTER 5

SOFTWARE REQUIREMENT SPECIFICATION

With reference to Index, SRS for all the fields should be specified according to the Project

CHAPTER 6

DETAILED DESIGN

6.1 INTRODUCTION

This document specifies the design that is used to solve the problem of Product.

6.2 ARCHITECTURAL DESIGN

A description of the program architecture is presented with architectural design.

6.3 DATA DESIGN

A description of all data structures including internal, global, and temporary data structures, data base design(tables), file formats.

6.3.1 Internal software data structure

Data structures that are passed among components of the software are described.

6.3.2 Global data structure

Data structure that are available to major portions of the architecture are described.

6.3.3 Temporary data structure

Files created for interim use are described.

6.3.4 Database description

Database(s)/Files created/used as part of the application is(are)described.

6.4 UML DIAGRAMS

All UML diagrams should be represented here

CHAPTER 7
OTHER SPECIFICATIONS

7.1 Advantages:

Project Advantages should be specified(12 sentence case)

7.2 Limitations:

Project Limitations should be specified(12 sentence case)

7.3 Applications:

Project Applications should be specified(12 sentence case)

CHAPTER 8
SUMMARY

One page summary of the project should be written.

ANNEXURE A

PROJECT PLANNER

Using planner or a like project management tool.

ANNEXURE B

REVIEWERS COMMENTS OF PAPER SUBMITTED

(At-least one technical paper must be submitted in Term-I on the project design in the conferences/workshops in IITs, Central Universities or Conferences or equivalent International Conferences Sponsored by IEEE/ACM)

1. Paper Title:
2. Name of the Conference/Journal where paper submitted:
3. Paper accepted/rejected:
4. Review comments by reviewer:
5. Corrective actions if any:

ANNEXURE C

PLAGIARISM REPORT

Any online tool can be used to check the Plagiarism of report.

REFERENCES

REFERENCES SHOULD BE INCLUDED IN IEEE FORMAT